

Primer - Travers AI Datacenter

Note prior to reading: This primer is provided as for information only to explain some of the concepts generally related to the design of the AI Datacenter that was proposed to the Vulcan County Council as part of the Public Hearing on April 1, 2026. This does not represent a detailed assessment of the design as the documents provided for the hearing are preliminary in nature. Detailed Engineering will need to be completed by the developer prior to any true or further assessment of the final design.

The document below has 2 sections:

1. A quick reference section (1 page) for very basic answers to some questions
2. An Appendix with more detailed answers to the same questions along with a Challenges/Mitigations section of what can be encountered in industrial facilities such as these.

Quick Reference

What is a datacenter? A facility that houses computer processing and data storage equipment.

What is different about an AI (Artificial Intelligence) datacenter? Due to the types of calculations required, these datacenters require higher computer processing power.

Why do AI datacenters require so much power? Powering the computer gear required for the processing of this data and the cooling systems that support that equipment.

Why do some AI datacenters require so much water? With the higher heat produced by the computer gear, evaporative cooling is generally used during hotter months of the year. The evaporated water needs to be replaced in the system.

What does “closed loop” cooling mean? Closed loop refers to a system that circulates around without no new medium (in this case liquid) coming in or out. Generally liquid touching the computer components is a closed loop, however evaporative cooling is an open loop as the evaporated water does not return.

Why does the AI datacenter need so much water continuously, don't they just fill up the system once? The evaporated water does not return and would have to be replenished.

Potential water use proposed? For 2GW the water license requested, based on the presentation from Palliser Grid was for 5500 acre-ft/year. More explanation in the appendix related to acre-ft and this topic.

What is the Wastewater Treatment Plant for that was shown in the proposal? Water brought in from the reservoir as well as recycled in the cooling system (the water that didn't evaporate) needs to meet certain design conditions to ensure the system continues to work properly.

Why all the batteries on the site? For power to be stored from renewable sources or the Alberta grid when that power is available.

Backup Power vs Power Generation The proposal lists diesel backup generators being required for the site that covers the full load, which will require significant infrastructure. Based on comments from the Alberta Government related to new AI datacenters, it is likely that the facility would be required to generate its own power on site; likely with a turbine generator(s) for full time power needs.

Powerlines For the full 2GW facility it could require a 500kV line depending on final design. Initial phase, again dependent on design and power generation, could be a 240kV line.

Challenges and Potential Mitigation Overview: Noise, Light Pollution, Egress, Fires, Air Emissions, Impacts to Wildlife/Environment and Construction Traffic – See Appendix

Appendix – Detailed Overview

Acronyms

AUC – Alberta Utilities Commission

[Home - AUC](#)

The Alberta Utilities Commission regulates the utilities sector, natural gas and electricity markets to protect social, economic and environmental interests of Alberta where competitive market forces do not

AESO – Alberta Electrical System Operator

[Home » AESO](#)

AESO mandate is to provide for the safe, reliable and economic operation of the interconnected electric system and to promote a fair, efficient and openly competitive electricity market. They are the planners and direct the safe operation of the Alberta grid.

Current load and Generation in the province if of interest to you:

ets.aeso.ca/ets_web/ip/Market/Reports/CSDReportServlet

BRID – Bow River Irrigation District

[Bow River Irrigation District](#)

BRID owns and operates several hundred kilometres of earth canals and water pipelines, as well as several reservoirs. This infrastructure is used to provide irrigation and domestic water for farming, industry, wildlife, and communities within the district's boundaries.

Travers is one of their reservoirs.

BYOP – Bring Your Own Power: the requirement to put in your own power generation facilities to accompany your development project that needs it.

What is a datacenter?

Facility that contains computer systems for storing and processing information. Generally contain servers that use CPUs (Computer Processing Units) and data storage like the ones you have in your computer at home (though more powerful) and the supporting infrastructure to power and cool the computer systems. With the work your computer does at home it is usually sufficient to have a fan on the processor and power supply cooling it by simply running air over cooling fins. These facilities can run applications, store websites, run basic searches and store large amounts of information online for us to access. (among other things)

What is different about an AI (Artificial Intelligence) datacenter?

Artificial intelligence models/calculations need a lot of processing power to work and generally rely more heavily on GPU (Graphics Processing Units) that tend to run these types of calculations better. You may have a separate graphics card in your home computer that would have this type of processor on it, but in an AI datacenter this would be on a commercial scale. Due to the heavy calculations required, these computer systems create a great deal more heat than a typical datacenter and may require liquid cooling of the components to operate properly. The processor transfers its heat to a liquid to “pull” the heat away from the computer components similar to how engine coolant in your car pulls heat away from the engine and then uses the radiator to transfer the heat into the air. (more about that below on the different type of coolers) These facilities are used for training AI models to be smarter, research and for things like ChatGPT or Google AI searches that you have likely seen recently. (among other things)

Why do AI datacenters require so much power?

Think of your computer at home only much more powerful (from a performance perspective) and then there could be 100s or 1000s of these units that are working in a single facility 24 hours a day, 7 days a week dependent on the size of the facility. The electrical load is to power these computer servers as well as for the pumps/fans and other equipment to cool them, which is significant.

Some Context for scale: The first proposed phase for the facility at Travers is 500MW. In the proposal for the development it notes that the facility could grow to 2000MW (2GW) over a longer period of time.

The Travers Solar farm (one of the largest solar farms in North America) has an installed capacity of 465MW, though keep in mind that is what it is ultimately capable of in ideal sunny conditions, not necessarily what it can continuously produce. (or at night)

[Travers Solar Project - Alberta Major Projects](#)

The largest single power generating facility in Alberta is the Shepard Generating Station at a maximum capacity of 868MW (0.868GW) that supplies power into the Alberta grid. Photos on the website below as well as more information on the facility.

[The Shepard Energy Centre | Electricity Canada](#)

Why do some AI datacenters require so much water?

Cooling. Where a regular datacenter may be able to use air cooling fans and air conditioning of the buildings to keep the equipment cool enough, an AI datacenter needs to be able to deal with much higher heat loads and therefore typically uses liquid cooling of the computer equipment. Liquid is able to pull away and remove heat faster and more effectively than air. There are different ways of cooling the liquid once it is heated up by the computer equipment, but one of the more common ways for AI datacenters is a combination of air cooling and evaporative cooling.

Air cooling (referred to as “dry cooling” in the Palliser Grid public hearing submission) is similar to the radiator in your car that takes heat from the engine into the coolant fluid and then removes the heat by having air pass through the radiator, air absorbs the heat, which cools the fluid and then goes back into the engine again for another cycle.

Evaporative cooling uses water being evaporated to create a cooling effect on the cooling fluid to remove the heat from the process. This is a similar concept to why your body sweats when it is trying to cool down: the sweat, being mostly water, evaporating creates a cooling effect on your skin as it takes energy (in this case heat) away to evaporate the water.

Air cooling works well when the air around the radiator is colder than the fluid it is trying to cool, like in the fall or winter. Due to the larger temperature difference outdoors, it is easier for the heat to be removed similar to what you would experience if you have your front door open in winter – heat escapes easily. If the temperature of the air is high, like in the summer, you won’t be able to pull as much heat out of the fluid. That is where you will have to rely on something different to cool your coolant fluid. For AI datacenters, evaporative cooling commonly comes in to resolve this issue. The reason that evaporative cooling is used is that it uses less power than mechanical refrigeration. Mechanical refrigeration

uses the same principles like you would have in your house Air Conditioning or your refrigerator and is sometimes referred to in the process as a “chiller”. The tradeoff for evaporative cooling is that water evaporates into the air and doesn’t come back into the system which is why more water then has to be added back into that part of the system.

What does “closed loop” cooling mean?

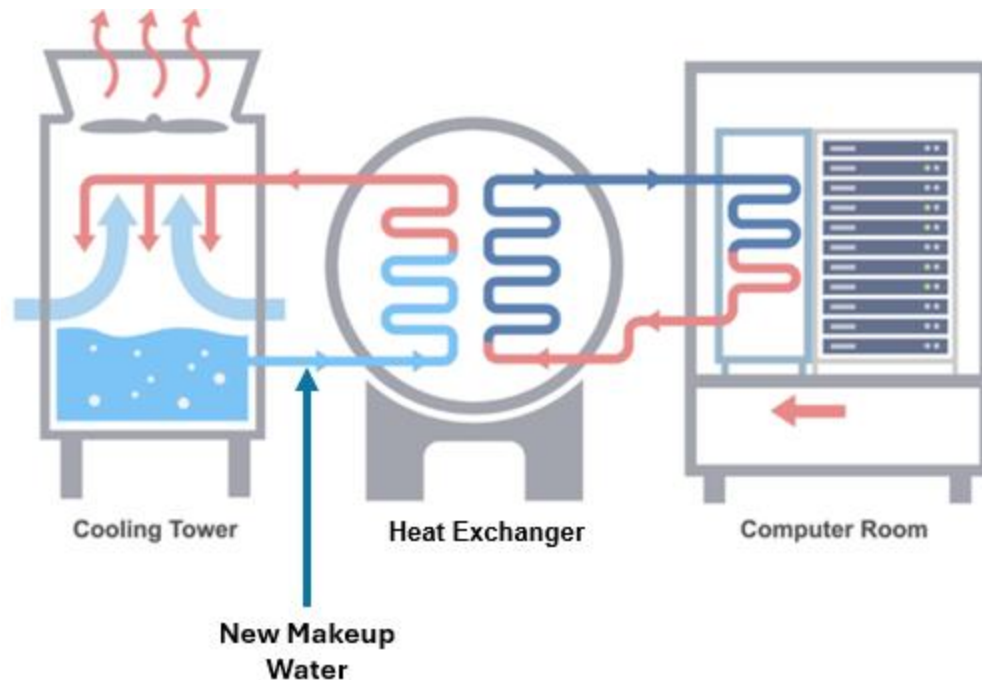
Closed loop refers to a system that circulates around without no new medium (in this case liquid) coming in or out. You have to be a little careful with the terminology here and have to take the whole cooling system into account to truly call it closed.

In AI datacenters that use evaporative cooling there is generally a closed loop side (computer side cooling loop) and an open loop side. (evaporative cooling side)

There are newer designs that replace the evaporative cooling side with closed loop air cooling or mechanical refrigeration, but as in the above section they are sometimes used in combination with evaporative cooling for areas that have very hot outdoor conditions. In the winter, a hybrid system like this can take advantage of air cooling to reduce the need for evaporative cooling that uses water.

Why does the AI datacenter need so much water continuously, don’t they just fill up the system once?

Not exactly, there are two different pipe “systems” that are circulating fluid. One part of the system is the fluid that comes in contact with the hot computer components and then travels to a heat exchanger that “moves” the heat into a different fluid. The two fluids don’t actually mix or directly touch each other; the heat is transferred through the metal in contact with both fluids. The other fluid would be water that would be pulled and treated from the Travers reservoir and would be pumped through the other side of the heat exchanger. This fluid would pick up the heat and then go to the evaporative or air-cooling units. The water in this evaporative cooling side of the system would have a portion of the water evaporated into the air to create the cooling effect required to cool the liquid. The evaporated water (which goes into the air) needs to be continuously replaced, hence the water usage that would be driven by how much cooling is required and the outside air temperature. Note that there are variations of the process above and how they are designed, but the basic principle of using evaporated water to cool remains the same. The graphic below shows the two loops and where new water would come into the process referred to as make up water.



Graphic Used and Modified to Show Makeup Water and Heat Exchanger from: [Managing AI's Heat: Advanced Liquid Cooling Systems and the Instrumentation That Supports Them](#)

A example of a manufacturer of cooling towers for some further info: (this is not necessarily what may be used, example only for general understanding)

[What is a Cooling Tower | Baltimore Aircoil](#)

Potential water use proposed

According to the Palliser Grid presentation at the public hearing, BRID appears to have granted a water licence for 5500 acre-ft per year for the Travers AI datacenter facility. This was noted to be the full built out at 2GW, not the initial phase. Assuming the first phase remains 500MW and **if** that water use was simply scaled, that would put the first phase at ~1375 acre-ft/year usage. As noted in other sections, this will be technology and seasonally dependent. If different cooling technology is used, this number would need to be revisited.

Context for scale: the spillway during the summer brings in 3000-3500 acre-ft in 1 to 1.5 days to the travers reservoir when it is flowing. This is an average, not a min or max.

More information is available on the irrigation setup on the BRID website:

[Bow River Irrigation District](#)

What is an acre-ft of water use?

This is just a different way to express volume: the amount of water that would fill one acre of land a foot high. In metric: 1233 cubic meters (1,233,480 liters) is equivalent to 1 acre foot. The “per day” or “per year” then allows you to understand the flowrate based on what period of time you are measuring the amount of water.

What is the Wastewater Treatment Plant for that was shown in the proposal?

Not all the water evaporates in the evaporative cooling cycle and is then recycled through the system prior to trying to evaporate it again. A wastewater treatment plant is generally in the cycle to ensure the quality of the water returning into the system meets certain design conditions to ensure the system continues to work properly. The system may be used to treat the incoming water from the Travers reservoir as well. Eventually some water will have to be disposed of when it can no longer be economically treated to be recycled. Though not noted in the proposal, generally this is trucked or conveyed through a pipeline to a disposal well or facility. It was noted verbally by the developer that no water would be returned to the Travers Reservoir from the AI facility.

Why all the batteries on the site?

If the goal is to use a lot of the renewable energy in the surrounding area it makes sense to try and store as much energy as possible when the sun is shining and/or the wind is blowing. It can then be used when those sources aren’t delivering power. Alternatively, power can be pulled in from the Alberta grid at low usage times such as at night to store for use during the day.

Backup Power vs Power Generation

The proposal to Vulcan County notes Backup Power in the form of diesel generator(s) capable of maintaining the full load of the facility for 72 hrs. Backup units are generally not intended to run outside of emergency requirements and monthly testing to ensure they work. Taking care of the full load of the facility for 72 hrs requires significant infrastructure

at 500MW. Large industrial diesel generators generally range between 1MW to 4MW each which would require a large number of units to meet the demand requirements and would likely be used in combination with the energy from the battery storage.

Example of one manufacturer:

[Diesel Generator Sets](#)

The proposal also listed 5-million-liters of onsite diesel storage for the emergency generators. A picture from a project of the same volume is shown below. There is a person located by the valve at the front of the tank for some reference of size inside the red circle.



Courtesy of: [Project: Avenell – 5 Million Litre Diesel Tank - Sun Engineering](#)

The original proposal doesn't list full time power generation on site, however based on reports from the Alberta government, they will likely be requiring new AI datacenters to "Bring Your Own Power" (BYOP) as there is not enough power in the Alberta grid to meet the current and potential new load from predicted AI datacenters. The unit(s) would run continuously. This power generation would likely come from a turbine generator(s) that runs off Natural Gas or Liquefied Petroleum gas. Note this is only speculation given the size of the electrical load required and the type of equipment big enough to supply it. With the demand for turbines being high, driven somewhat by AI datacenters builds, the timeline for delivery of turbines has increased significantly vs typical demand.

[Artificial Intelligence Data Centres Strategy | Alberta.ca](#)

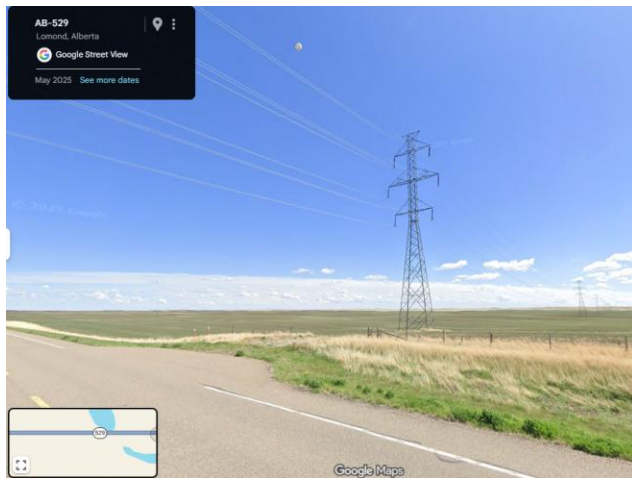
[Alberta: Power to compute - RBC](#)

Powerlines

New powerlines were proposed to the new AI datacenter site in the proposal to Vulcan County for grid connected power and also to likely try and use the renewal power in the area. (If that is contractually possible is not known) Given the expected future electrical requirements stated as 2GW it could be up to a 500kV transmission line required for the load dependent on final design. The facility would require a substation which is where the transmission lines would come into breakers and transformers to supply or allow for export of power.

Context for scale: the transmission lines that cross Hwy 529 east of the Travers Reservoir close to the solar farm is an example of a 240kV line. 240kV lines would generally be good for ~500MW. As the design is not finalized, it is not completely clear what would be applied for with the AUC. A 500kV line is a larger structure than a 240kV line. Examples of both types of lines are pulled from google maps to give you an idea of scale.

240kV Line East of Little Bow Resort. Courtesy of Google Maps.



Example of 500kV line (Edmonton beside Antony Henday Dr west of Hwy 1) – One of the structures, not both sets of structures. Courtesy of Google Maps.



Travers Solar Substation. Courtesy of Google Maps Images of Travers Solar Farm taken by Corbin Mortin.



Challenges and Potential Mitigations

This section comments on general challenges with industrial equipment related to AI datacenters and info on some mitigation strategies that are **generally** used in industrial facilities. NOTE: This only gives an overview of some technologies that exist, NOT that they are part of the current facility plan nor to express that they would be able to completely mitigate the issues presented.

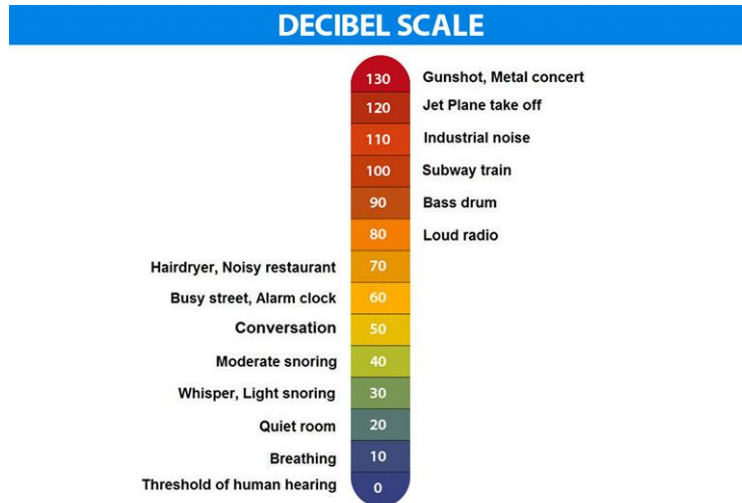
Noise

There would not be a great deal of noise generated from the actual computer equipment that would be located inside a building. The noise would predominately come from any power generation or fans that were part of the cooling process and other mechanical equipment such as pumps. The pumps may be contained in buildings, however the fans would be outdoors as well as any exhaust stacks from power generation.

Mitigations you may see: Berms or walls to attempt to block or dampen the sound. Also sound dampening equipment such as mufflers on power generating equipment or baffled enclosures around fans. An Engineering noise study would be required with particular attention to low frequency noise.

How is noise measured?

Measured in Decibels. (Db) This measurement is not linear like turning up your stereo, it is logarithmic so a number 2x as big isn't twice as "loud". See below for some examples related to decibel levels and common sounds.



Graphic courtesy of: [Generator Sound Levels Explained: Noise Reduction | Generator Power](#)

Light Pollution

Lights around the facility adding to a predominantly dark area. Light can sometimes also be reflected off any condensed water clouds rising into the sky created by cooling or turbine exhaust dependent on outdoor environmental conditions.

Mitigations you may see: There are directed lighting options to minimize upward facing lighting. Engineering study would be required.

Egress

Egress refers to the path of leaving a place or, related to industrial facilities, ensuring that if there is an emergency event that there are routes for workers and/or the public to exit safely. Studies would have to be completed to ensure that there is safe route for people to exit the area, which may require new routes/roads or upgrades to existing roads to be completed.

Fires

Any industrial facility would need to put together a proper design and plan for fire protection that meets the required codes they fall under and the requirements of the municipality. This may require water deluge (like fire spray nozzles you would see in

commercial building roofs) or foam systems to smother a fire or CO₂ systems on equipment like turbines or computer equipment that starve the fire of oxygen that it needs to continue to burn. The type of equipment and types of fuels on site will guide which systems are used and may also require a “fire water loop” so that emergency services have easier access to a water supply. This loop is essentially a water line with hydrants attached to a pumping system that is setup in a way that is unlikely to be compromised during a fire. Engagement with local fire departments would also be important to ensure they are equipped and trained in the types of fires they may encounter such as with the batteries that are proposed to be on site. In some cases, specialized firefighting equipment or personnel trained to fight those types of fires may need to be called upon by the owner of the facility if local resources aren’t specifically trained for that given event.

Air Emissions

Air emissions areas of focus would be related to power generation equipment from the turbine(s) or diesel generators for emergency backup power that would contain NO_x, particulates and CO₂ in the exhaust streams. Small water droplets containing dissolved solids or treatment chemicals would be the other focus related to the cooling towers depending on how they are configured.

Mitigations you may see: The gas turbine can be designed to include lean burn technologies to reduce NO_x emissions. More elaborate selective catalytic reduction technologies can be applied to decrease NO_x, however they carry some other downsides as well. There are filter technologies related to small water droplet removal and controls that can be applied to cooling towers. Carbon capture pipelines related to CO₂ are not currently planned for this area of the province. Engineering studies would be required to ensure that the equipment meets Alberta Environment and Canadian Industrial standards.

Impacts to Wildlife/Environment

Given the proximity to the Travers reservoir and some of the sensitive areas next to the water, a wildlife and Environmental assessment would be required. This is generally a requirement with any AUC application.

Construction Traffic

Generally the County or municipalities will require a developer to complete and submit a traffic study as part of the development process that will contain mitigation to traffic issues and congestion.